BlueBoat Configurable Simulator

Design, assumptions, configuration, and verification

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The simulator supplies a repeatable environment for developing boat controllers. It separates hull motion, M200 propulsion, command execution, and sensor delivery so that each assumption can be tested independently. This report documents the implemented equations and information flow, gives the complete flag reference, and explains the saved model comparisons.

- Begin with a local linear plant around a declared operating speed.
  - Add nonlinear hull and propulsion effects, then motor response, timing, sensing, and disturbances.
    - Compare identical commands and initial conditions before attributing a change to the selected feature.
- Keep simulated motion separate from controller observations.
  - The internal state supplies numerical integration and scoring; measurement delivery supplies the future feedback interface.
    - An explicit perfect-state switch supports ideal controller tests. Ordinary sensor tests require an estimator.
- Distinguish implementation verification from boat validation.
  - Automated checks establish consistency, causality, bounds, and repeatability.
    - Measured boat logs establish predictive accuracy. Added model detail does not by itself establish improvement.

> The saved demonstration uses an open-loop command program. The companion controller report adds proportional-integral-derivative control, linear quadratic regulation, pole placement, and state estimation through the same motor-percentage interface.

The physical demonstration follows baseline revision `ea307b3`, with its commands now expressed as signed motor percentages. The companion controller report supplies the subsequently implemented feedback study. Its diagrams use a dedicated navy-and-teal language, and its quantitative charts use standard plot colors. Every figure is authored in Typst with CeTZ or Lilaq, with inspectable source data retained beside the chapters.

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1 Hull dynamics

The simulator advances a planar hull state from body-frame velocities and two independently commanded propulsors. This chapter defines its coordinate convention, derives the implemented equations, and relates the linear baseline to the nonlinear hull model.

1.1 Frames and state

The inertial frame uses east E and north N position coordinates. The body frame uses forward x_b and port/left y_b axes, with heading ψ measured counterclockwise from east. Consequently, a zero-heading boat points east, while a positive heading rotates its forward axis toward north. Surge u, sway v, and yaw rate r are body-frame velocities. The six-state hull vector is $\eta = (E, N, \psi)$ and $\nu = (u, v, r)$.

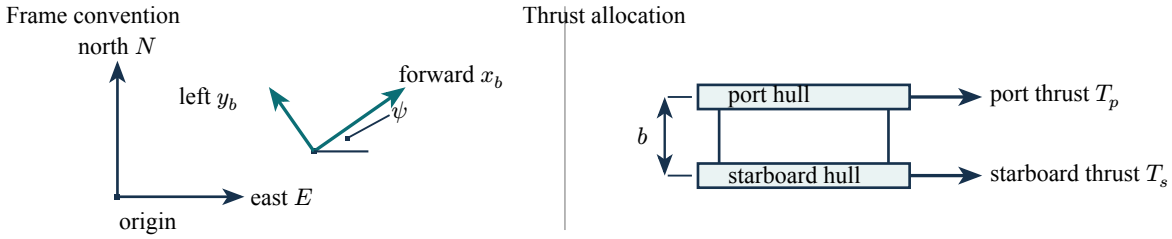


Figure 1: Body forward and left axes rotate with heading, while east and north remain fixed.

That convention fixes both the position rotation and the sign of differential thrust. The exact kinematic relation is

$$\dot{E} = \cos(\psi)u - \sin(\psi)v, \quad \dot{N} = \sin(\psi)u + \cos(\psi)v, \quad \dot{\psi} = r. \tag{1}$$

In matrix form, $\dot{\eta} = J(\psi)\nu$, where the planar rotation block is

$$J(\psi) = \begin{pmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{2}$$

The two frames keep position and velocity descriptions consistent as the craft turns.

- The earth frame records where the hull is, while the body frame records how it moves relative to its own axes.
  - Surge is forward and sway is left, so the body axes rotate together with heading.
  - Applying $J(\psi)$ resolves these velocities into east and north components at every step.
- Heading is positive counterclockwise from east.
  - At $\psi = 0$, forward points east; at positive ψ, forward has a northward component.
  - This sign convention makes positive yaw and a larger starboard thrust produce a counterclockwise turn.

1.2 Nonlinear planar force balance

The force model follows the standard marine-craft decomposition of inertia, velocity coupling, damping, and applied effort. Here each term is specified by the implementation and configuration; the general modeling framework is consistent with Fossen's marine craft formulation [1].

The effective diagonal inertia is $M = \operatorname{diag}(m_x, m_y, I_z)$. The nominal rigid-body yaw inertia is I, while I_z denotes its effective value after the optional added-inertia contribution. With rigid-body mass m and added-mass flag $a \in \{0, 1\}$, code sets

$$m_x = m(1 + 0.10a), \quad m_y = m(1 + 0.50a), \quad I_z = I(1 + 0.20a). \tag{3}$$

The implemented velocity-coupling vector added on the right-hand side is $c(\nu) = \big(m_y vr, -m_x ur, (m_x - m_y)uv\big)^T$. Under the conventional left-hand-side form $M\dot{\nu} + C(\nu)\nu = \tau$, this is $c(\nu) = -C(\nu)\nu$. Its structure exchanges energy between surge, sway, and yaw without adding or removing total kinetic energy, since

$$\nu^T c(\nu) = u m_y vr - v m_x ur + r(m_x - m_y)uv = 0. \tag{4}$$

This cancellation is a useful structural check on the cross-coupling signs. The model's body-frame equations are

$$m_x \dot{u} = T_p + T_s + m_y vr - D_{u(u)} + \tau_u. \tag{5}$$

$$m_y \dot{v} = -m_x ur - D_{v(v)} + \tau_v. \tag{6}$$

$$I_z \dot{r} = \frac{b}{2}(T_s - T_p) + (m_x - m_y)uv - D_{r(r)} + \tau_r. \tag{7}$$

Here T_p and T_s are port and starboard thrust in newtons, b is their lateral separation, and (τ_u, τ_v, τ_r) denotes environmental surge force, sway force, and yaw moment in body coordinates. Positive differential effort means starboard thrust exceeds port thrust and therefore produces positive, counterclockwise yaw.

The drag laws are signed so that they oppose velocity. In nonlinear mode they are

$$D_{u(u)} = d_1 u + d_2 u|u|, \quad D_{v(v)} = d_v v + k_v v\,|v|, \quad D_{r(r)} = d_r r + k_r r\,|r|. \tag{8}$$

Configuration defaults set $k_v = k_r = 0$. Thus default lateral and yaw drag are linear, while surge drag combines linear and quadratic terms. The quadratic surge term captures increasing resistance with speed while retaining the correct sign in reverse.

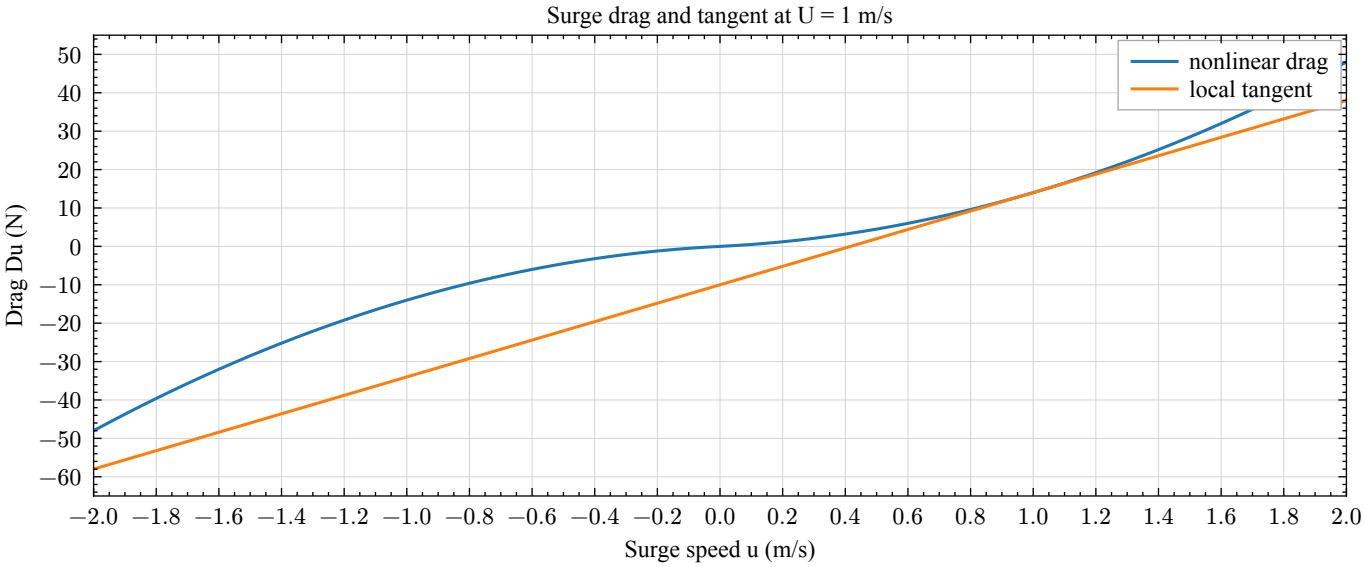


Figure 2: The straight-line approximation matches nonlinear surge drag and slope at the one-meter-per-second operating point.

1.3 Hull and coefficient values

The numeric hull begins with the 1.2 m length and 0.93 m beam of the configured platform. The implementation computes a 21.9 kg rigid-body mass as 14.5 kg bare hull plus 2.4 kg for two batteries and a 5.0 kg equipment allowance. The nominal yaw inertia uses a uniform rectangular-envelope approximation, $I = \frac{m(L^2+B^2)}{12} = 4.2064425$ kg m². This is a geometry-based modeling choice, not a measured mass distribution. Thruster spacing is configured as $b = 0.75$ m.

The configured values combine platform dimensions, mass accounting, and assumed resistance coefficients.

- Platform geometry and mass determine the rigid-body terms.
  - The 1.2 m length, 0.93 m beam, 14.5 kg bare hull, and 2.4 kg battery total are platform specifications; the 5.0 kg equipment allowance is a configuration assumption.
  - Together they produce $m = 21.9$ kg; the 0.75 m thruster separation is also a configuration assumption.
- The inertia proxy converts the platform envelope into yaw resistance.
  - The model uses $I = \frac{m(L^2+B^2)}{12} = 4.2064425$ kg m², giving $I_z = I$ when added mass is off.
  - With added mass on, the configured fractions give $m_x = 24.09$ kg, $m_y = 32.85$ kg, and $I_z = 5.047731$ kg m².
- Drag coefficients set the resistance used by the hull equations.
  - The assumed surge terms are $d_1 = 4$ N s/m and $d_2 = 10$ N s²/m²; sway and yaw use $d_v = 35$ N s/m and $d_r = 12$ N m s.
  - The configuration defaults omit quadratic sway and yaw drag, so $k_v = k_r = 0$.

Thrust allocation maps the two motors to generalized effort as

$$\tau_T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ -\frac{b}{2} & \frac{b}{2} \end{pmatrix} \begin{pmatrix} T_p \\ T_s \end{pmatrix}. \tag{9}$$

At $b = 0.75$ m, equal thrust changes surge without yaw, while an equal-and-opposite change creates yaw with no net surge. Installed thrust is based on a signed static motor curve scaled so the pair reaches the platform's stated 8.2 kgf forward maximum. Reverse scaling and between-point values remain assumptions in that map; the hull force equations themselves consume thrust in newtons.

1.4 Local linear model and feature switches

The initial configuration holds $U = 1$ m/s and $\psi_0 = 0$. The nominal position evolves as $E_0(t) = Ut$ and $N_0(t) = 0$, so $\tilde{E} = E - E_0(t)$ and $\tilde{N} = N - N_0(t)$ describe departures from that moving reference. For small velocity and heading perturbations $\tilde{u}, \tilde{v}, \tilde{r}, \tilde{\psi}$, with trim thrust balancing drag, the linearized equations are

$$\dot{\tilde{E}} = \tilde{u}. \tag{10}$$

$$\dot{\tilde{N}} = \tilde{v} + U\tilde{\psi}. \tag{11}$$

$$\dot{\tilde{\psi}} = \tilde{r}. \tag{12}$$

$$\dot{\tilde{u}} = -\left(\frac{d_1 + 2d_2U}{m_x}\right)\tilde{u} + \frac{\widetilde{T_p} + \tilde{T}_s}{m_x}. \tag{13}$$

$$\dot{\tilde{v}} = -\left(\frac{d_v}{m_y}\right)\tilde{v} - \left(m_x\frac{U}{m_y}\right)\tilde{r}. \tag{14}$$

$$\dot{\tilde{r}} = \left(\left(m_x - m_y\right)\frac{U}{I_z}\right)\tilde{v} - \left(\frac{d_r}{I_z}\right)\tilde{r} + \frac{b\left(\tilde{T}_s - \widetilde{T_p}\right)}{2I_z}. \tag{15}$$

These equations give a readable state-space model $\dot{\tilde{x}} = A\tilde{x} + B\tilde{T}$ for $\tilde{x} = \left(\tilde{E}, \tilde{N}, \tilde{\psi}, \tilde{u}, \tilde{v}, \tilde{r}\right)$ and $\tilde{T} = \left(\widetilde{T_p}, \tilde{T}_s\right)$. For the default no-added-mass coefficients, the nonzero entries of A are $A_{\{E,u\}} = 1$, $A_{\{N,v\}} = 1$, $A_{\{N,\psi\}} = U = 1$ m/s, $A_{\{\psi,r\}} = 1$, $A_{\{u,u\}} = -1.0959$ s^{-1}, and $A_{\{v,v\}} = -1.5982$ s^{-1}, $A_{\{v,r\}} = -1.0000$ m/s, $A_{\{r,r\}} = -2.8523$ s^{-1}. The input matrix rows for u are $(0.04566, 0.04566)$ kg^{-1}; rows for r are $(-0.08915, 0.08915)$ with units kg^{-1} m^{-1}. All other entries of A are zero.

The feature flags select whether this local approximation or the corresponding nonlinear relation supplies each component.

- `nonlinear_kinematics` switches from the local position approximation to exact rotation at the current heading. At heading zero it uses lateral velocity $v + U(\psi - \psi_0)$, retaining the first-order effect of heading on cross-track motion.
  - Near $U = 1$ m/s and small heading error, this agrees to first order with exact rotation. Large heading changes require the nonlinear rotation.
- `nonlinear_coupling` replaces the constant-speed linear terms $\left(-m_xUr, \left(m_x - m_y\right)Uv\right)$ with the full state-dependent terms $\left(m_yvr, -m_xur, \left(m_x - m_y\right)uv\right)$.
  - The local coupling evaluates sway/yaw cross terms at trim speed U and discards nonlinear state products.
- `nonlinear_drag` replaces the tangent drag law around U with signed quadratic surge drag and configured sway/yaw quadratic terms.
  - In surge, linear mode evaluates $D_{u(U)} + (d_1 + 2d_2|U|)(u - U)$. It is accurate near the trim speed and diverges increasingly from the nonlinear curve farther away.
- `added_mass` scales the diagonal effective inertia by the configured fractions and changes the coupling coefficients through m_x, m_y, I_z.

Accordingly, the baseline is a local model for small perturbations around its selected speed and heading. Turning on the `01_hull` preset restores exact position rotation, nonlinear coupling, and nonlinear drag, while preserving the same mass and nominal coefficient choices. These equations describe the implemented hull dynamics; motor response, command timing, sensors, and disturbances are separate simulator layers.

2 Actuation and Command Timing

2.1 From motor percentage to thrust

The propulsion model starts from Blue Robotics' signed M200 static samples at 16 V, then scales both motors uniformly to match the BlueBoat's published 8.2 kgf forward rating. This joins a component curve to an installed-boat rating as a provisional model assumption. The product information supplies the 8.2 kgf figure, and the retained technical-reference samples supply the M200 curve [2], [3].

Signed motor command a runs from −100% full reverse through 0% neutral to +100% full forward. It denotes normalized drive request, rather than percent thrust or power. The adapter converts this to the device pulse width p before evaluating the retained map.

$$p(a) = (1500 + 4a)\ \mu\text{s}. \tag{16}$$

The force assigned to a motor follows

$$F_{\text{inst}(p)} = sF_{16(p)}, \quad s = (8.2\ \text{kgf})\frac{9.80665\frac{\text{N}}{\text{kgf}}}{2\max_p F_{16(p)}}. \tag{17}$$

Here F_{16} is the piecewise-linear interpolation of the retained 16 V samples, and $s = 0.72824$ is dimensionless. The result sets the positive limit to $40.207N$ per motor and the native reverse limit to $-20.068N$ per motor. The forward limit is anchored to the product rating. Uniform scaling of reverse and intermediate values is the model's installation assumption.

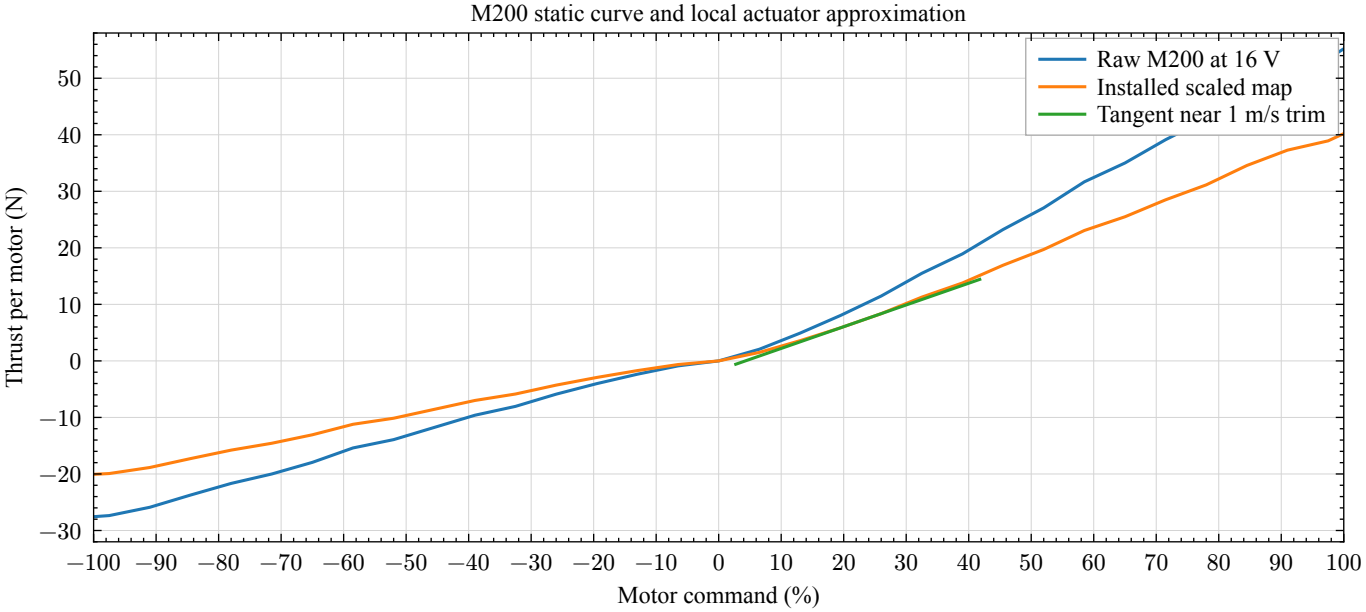


Figure 3: The installed map matches the pair's forward rating; the tangent is local to trim.

The simulator offers two actuator descriptions so the experiment can isolate nonlinear effects from a local approximation.

- The nonlinear lookup path uses the installed curve. Input percentages stay within −100% through +100%; the saturation flag also bounds force.
  - With neutral deadband enabled, commands from −6.25% through +6.25% produce zero thrust.
  - With reverse asymmetry enabled, commands below neutral use the signed reverse branch, retaining the lower reverse limit. When disabled, the model mirrors the forward branch into reverse and therefore removes the measured map asymmetry.
- The local model computes $F = F_0 + k_a(a - a_0)$ around a selected trim.
  - At 1 m/s, the standard linear and quadratic surge drag values require $7.0N$ per motor. The map gives $a_0 = 22.475$ percent, with local slope $k_a = 0.38455$ N per percentage point.
  - At 25%, both predict $7.971N$ because trim and 25% lie in the same piecewise-linear map segment. At 50%, the lookup gives $18.854N$ while the tangent extrapolation gives $17.585N$, a $1.269N$ difference.

The force path also allows a channel mismatch after percentage-to-force conversion. Port and starboard gains default to 0.95 and 1.05, then thrust bounds are reapplied if saturation is active. For example, at the plotted 50% command, the native installed lookup gives $18.85N$ before channel gains. The channel targets become $17.91N$ and $19.80N$, so equal percentage commands can create a $1.89N$ thrust difference and an unintended yaw moment.

2.2 Motor response after a command

Once an applied command sets a new target, the simulator evolves motor force with an optional first-order lag and optional force slew limit. With lag time constant τ and rate limit R, define error magnitude $m = |F_t - F_0|$, direction $q \in \{-1, 0, 1\}$ from current force toward target, and transition time $t_r = \frac{\max(m - R\tau, 0)}{R}$. The remaining error is

$$e(t) = \begin{cases} m - Rt & 0 \le t \le t_r \\ \min(m, R\tau)\exp\left(-\frac{t - t_r}{\tau}\right) & t > t_r \end{cases} \qquad F(t) = F_t - qe(t). \tag{18}$$

This combined law initially slews at up to R and then follows the first-order exponential tail. When only lag is enabled, the full response is exponential. When only the rate limit is enabled, force moves at constant rate until it reaches the target. With both features disabled, target force applies immediately.

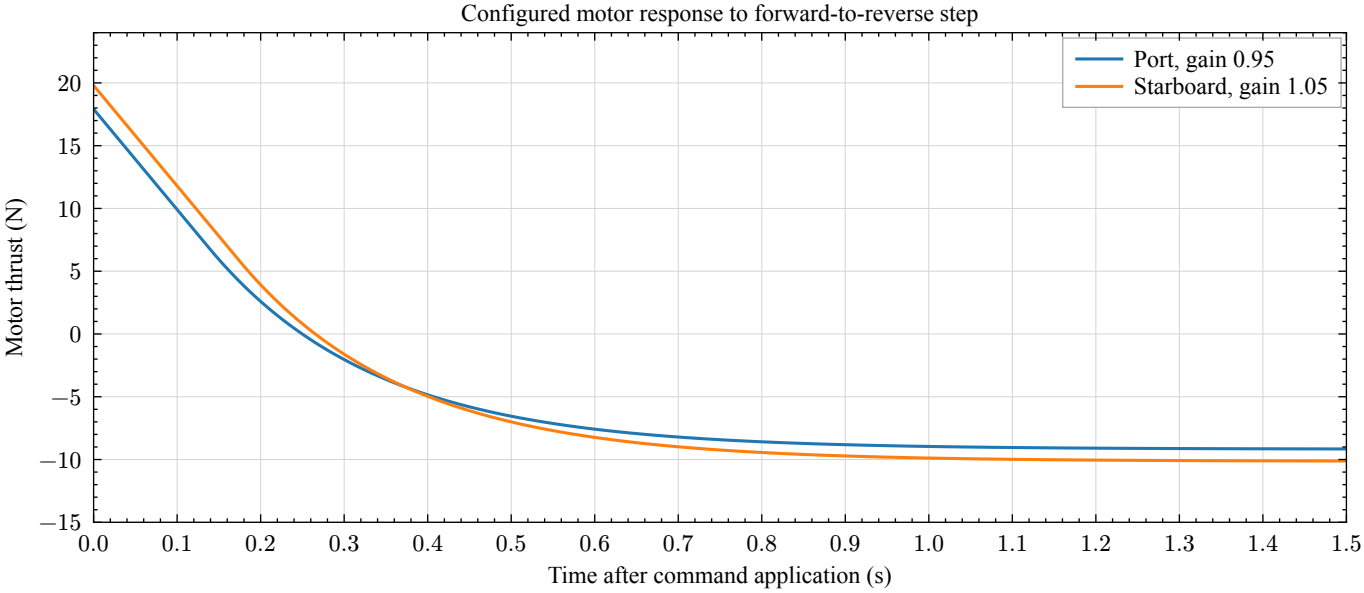


Figure 4: Motor force response to a +50% to −50% reversal after the target applies.

For this plotted reversal, each motor changes target from forward to reverse while its state starts at the corresponding 50% steady force. The port target moves from $17.91N$ to $-9.17N$, and the starboard target moves from $19.80N$ to $-10.14N$. With $\tau = 0.2s$ and $R = 80\frac{N}{s}$, the port error magnitude is $27.09N$ and the linear portion lasts about $0.139s$, followed by an exponential tail. The starboard linear portion lasts about $0.174s$. The delay from command decision to applied target is separate from these motor dynamics.

2.3 Command clock and delivery faults

When command sampling is enabled, the configured 20 ms period gives a 50 Hz decision rate. When command delay is enabled, each command then passes through three configured delay terms: 5 ms for computation, 10 ms for transport, and 20 ms of actuator dead time. Their nominal sum is 35 ms, or 1.75 command periods. Optional timing jitter adds a uniform value from zero to 3 ms per decision. Because delayed commands are queued as events, more than one command may be in flight.

- Sampling and delays affect when a new percentage pair can change the target.
  - A missed update creates no application event, so the previously applied percentage and force target remain in effect.
  - The default missed-command probability is zero; fault experiments can raise it.
- The optional watchdog monitors applied command freshness.
  - Its default timeout is 100 ms. Each applied command schedules a timeout, and newer applications invalidate older timeout events.
  - If the active timeout expires, both commands return to 0% and both force targets become zero. Enabled lag and slew limits still govern how actual force approaches zero.

The 50 Hz command period, delay values, jitter bound, motor lag, rate limit, mismatch, and watchdog timeout are configuration values for engineering experiments. They are not measurements of onboard timing. Orin timing is represented only by an assumed computation delay; this simulator does not emulate an Orin processor, operating system, or software scheduling load. The exposed boundary accepts paired motor percentages and produces simulated thrust. It does not implement the stock autopilot software.

> Interpret each delay as an explicit test input. For a command issued at $t = 1.000s$, the nominal application time is $1.035s$ before jitter, and the post-application force response then follows the selected motor model.

3 Sensor measurement models

The simulator schedules global navigation satellite system (GNSS), inertial, and stereo observations from one vessel truth state. Each stream has its own sampling period, seeded noise and dropout draws, and delivery event. The selected

hardware motivates these measurement channels, while the simulator equations below define exactly what its controller boundary receives [4], [5], [6].

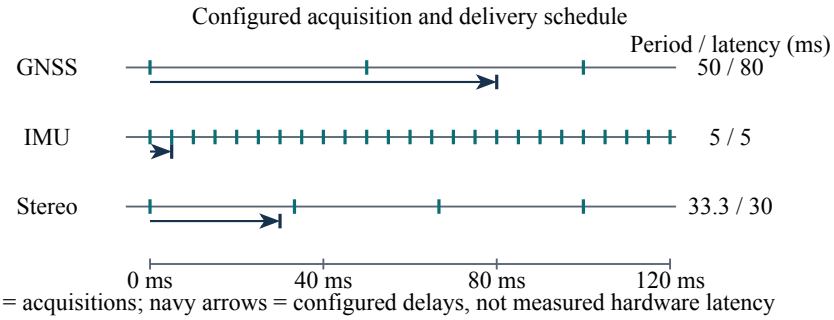


Figure 5: Acquisition ticks and configured delivery delays for the three independent sensor streams.

3.1 GNSS position, velocity, and heading

GNSS samples the vessel reference pose and body velocity, then rotates those quantities into the east-north frame. The configurable antenna location lies a distance ℓ forward or aft along the boat axis. Let (x, y, ψ) denote vessel reference position and heading, (u, v) body surge and sway speed, and r yaw rate. The antenna position and velocity are

$$x_a = x + \ell \cos\psi, \quad y_a = y + \ell \sin\psi, \tag{19}$$

$$v_{a,e} = u\cos\psi - v\sin\psi - \ell r \sin\psi, \quad v_{a,n} = u\sin\psi + v\cos\psi + \ell r \cos\psi. \tag{20}$$

- `antenna_lever_arm` selects $\ell =$ gnss_x_m; otherwise $\ell = 0$.
  - The default configured offset is -0.5 m, so the antenna is behind the reference point. Rotation contributes the tangential velocity terms proportional to ℓr.
  - GNSS heading is the simulated ψ with configured heading noise. The selected Mosaic-H is a dual-antenna heading receiver; a geometric baseline measurement can be represented as $\psi = \operatorname{atan2}(\Delta_N, \Delta_E)$, where (Δ_E, Δ_N) is the baseline vector between antennas. Baseline heading is not separately simulated.
- Position, velocity, and heading are one named measurement record.
  - Default standard deviations are 0.02 m per position coordinate, 0.03 m/s per earth-frame velocity coordinate, and 0.15 degrees for heading. The configuration calls its solution mode `RTK_FIXED`; it does not model solution-mode transitions.
  - Default position bias is zero. Configurable east and north biases and an optional antenna lever arm affect generated values when their flags are enabled.

3.2 Planar inertial measurements

The inertial channel reports yaw rate and body-frame specific acceleration at the vessel reference point. Its acceleration equations preserve the rotating body-frame kinematics, so directly exposing derivatives of u and v would omit the centripetal coupling terms:

$$\omega_z = r, \quad f_x = \dot{u} - vr, \quad f_y = \dot{v} + ur. \tag{21}$$

- The channel samples at 200 Hz by default and reports ω_z, f_x, and f_y.
  - Configured white-noise standard deviations are 0.002 rad/s for yaw rate and 0.02 m/s² for each acceleration component.
  - Yaw-rate bias defaults to 0.001 rad/s. Acceleration bias is zero in this model.
- The accelerometer values are computed from the planar physics derivative at acquisition time.
  - Gravity projection, roll and pitch, sensor-axis misalignment, scale factors, and temperature effects are not represented.
  - This output is suitable as an estimator input, but the simulator supplies no attitude filter or inertial integration.

3.3 Stereo target range and bearing

The camera abstraction uses a configured static world target and pinhole stereo geometry. It computes ideal target image center and disparity from truth, applies configured pixel noise, and converts those measurements to range and bearing. It does not produce image frames or perform detection, matching, or perception [6].

The default target is at east-north coordinates $(15, 3)$ m, while the initial vessel pose is $(0, 0, 0)$, so the initial target lies 15 m forward and 3 m to port. The resulting 15.3 m horizontal range is within the configured 20 m validity limit and projects inside the 1920 px image.

For world target (X_T, Y_T) and boat pose (x, y, ψ), define forward depth Z and lateral offset Y in the boat frame. The true horizontal range and bearing are $R = \sqrt{Z^2 + Y^2}$ and $\beta = \operatorname{atan2}(Y, Z)$. The image center coordinate c and stereo disparity d follow

$$c = c_0 - f_x \frac{Y}{Z}, \quad d = f_x \frac{B}{Z}, \tag{22}$$

where c_0 is half the image width, f_x is focal length in pixels, and B is stereo baseline. The noisy pixel coordinate produces $\hat{\beta} = \operatorname{atan}\left(\frac{c_0 - \hat{c}}{f_x}\right)$. The model recovers horizontal range as $\hat{R} = f_x \frac{B}{[\hat{d}\cos(\hat{\beta})]}$.

- A sample is geometrically valid only when depth and disparity are positive, the noisy target center lies within the configured image width, the target range is no greater than 20 m, and the configured occlusion window is inactive.
  - Random dropout can invalidate an otherwise valid observation. Invalid measurements retain their keys but carry missing values.
  - Default focal length is 700 px, baseline is 0.12 m, image width is 1920 px, center-coordinate noise is 2 px, and disparity noise is 0.5 px.
- Disparity uncertainty produces rapidly growing range uncertainty.
  - For a front-facing target with small bearing error, $R \approx Z = f_x \frac{B}{d}$. Linearizing this inverse-depth relation gives $\sigma_R \approx Z^2 \frac{\sigma_d}{f_x B}$.
  - The plotted curve uses configured f_x, B, and σ_d and spans the valid 20 m range for an on-axis target. It defines uncertainty as one standard deviation from disparity noise alone; pixel-center noise contributes additional bearing and horizontal-range error off the centerline.

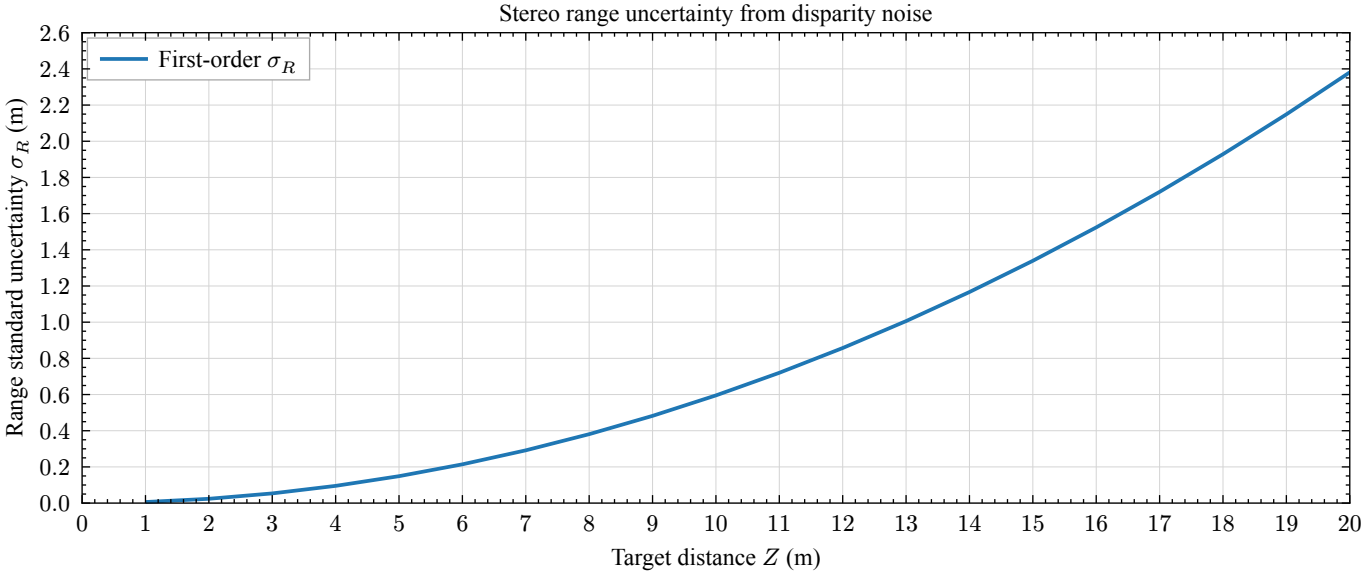


Figure 6: First-order stereo range standard uncertainty from the configured disparity noise and camera geometry.

At 10 m, the model predicts disparity $700\frac{0.12}{10} = 8.4$ px and first-order range standard uncertainty of about 0.60 m. At 20 m, disparity falls to 4.2 px while uncertainty grows to about 2.38 m. These values expose a geometry limit for downstream estimation; they are not output from image processing or a camera noise experiment.

3.4 Timing, uncertainty, and estimation boundary

The defaults below combine rates, delay settings, and noise scales from `Sensors`. Latency affects when a sample becomes visible, while uncertainty affects its value at acquisition. Each feature is gated independently by the corresponding experiment flags.

Stream	Rate	Latency	Default value uncertainty
GNSS	20 Hz	80 ms	0.02 m; 0.03 m/s; 0.15 deg
Inertial	200 Hz	5 ms	0.002 rad/s; 0.02 m/s²
Stereo	30 Hz	30 ms	0.5 px disparity; 2 px center

> Rates and latencies are configuration values. The standard deviations are injected only when sensor_noise is enabled; biases, delay, and random dropout likewise require their respective flags.

An acquisition record stores sample time, availability time, validity, delivered values, and the separate truth used to generate those values. The scheduler exposes only the latest delivered measurement per stream to a policy, so stale and delayed observations are distinguishable from the current state. Independent random streams let noise and dropout draws remain reproducible without coupling one sensor's samples to another's.

- Measurement history supports direct comparison against truth for scoring and debugging.
  - Invalid samples are recorded and delivered with missing values, preserving the timing of a lost update.
  - A pending sample is not available before its scheduled delivery time, even though its acquisition truth is already logged for evaluation.
- State estimation remains a consumer responsibility.
  - GNSS provides intermittent pose and earth-frame velocity, while the inertial stream supplies fast body motion cues and stereo supplies target-relative geometry.
  - A filter would need to handle asynchronous sample times, lever-arm corrections, noise and bias, and delayed or missing updates. The simulator does not combine these records into a fused state estimate.

4 Configuration and reproducible operation

The preceding chapters define the implemented physics and observations. Configuration selects which of those mechanisms participates in a run and supplies its numerical parameters. Flags choose a mechanism; parameters determine its strength, timing, and geometry.

4.1 Configuration hierarchy

- `Experiment` supplies the run name, duration, seed, operating point, initial state, and command scenario.
  - The default run lasts 12 s, starts at 1 m/s, and uses a 5 ms maximum integration step with 20 ms logging.
    - Sensor and command events split integration intervals at their actual times. The maximum step does not quantize a 35 ms command delay into an approximate event.
- Five nested groups describe the model.
  - `flags` contains 31 Boolean switches. `hull`, `actuator`, `environment`, and `sensors` contain numerical settings.
    - Disabled effects retain their parameter values so a paired experiment can enable them without silently changing another assumption.
- Full JSON templates define every preset.
  - `00_linear` through `07_uncertainty` contain resolved settings rather than hidden inheritance.
    - The configuration saved with a result is the final version after command-line overrides. Use that record to reproduce a run.

> A true flag can produce no numerical effect when its magnitude is zero. For example, enabled sensor dropout with probability zero retains all samples. Always inspect the resolved flag and its associated magnitude together.

4.2 Preset ladder

The presets collect related effects into a cumulative ladder. Each row adds the named mechanisms to the preceding configuration. Custom JSON files can select other valid combinations, and isolated comparison mode adds one group to the linear baseline with its required dependencies.

Preset	Additional effects
`00_linear`	Local hull and actuator linearizations; ideal command execution and enabled sensors. Perfect-state access is available.
`01_hull`	Actual heading rotation, nonlinear velocity coupling, and quadratic hull drag.
`02_m200`	Signed installed M200 lookup, saturation, neutral deadband, and reverse asymmetry.
`03_motor`	Motor force lag and force slew limits.
`04_timing`	Command sampling and compute, transport, and actuation delay.
`05_sensors`	Measurement noise, bias, latency, GNSS lever arm, and camera occlusion. Perfect-state access is disabled.
`06_disturbances`	Wind, correlated gusts, apparent wind, wind moment, and prescribed planar wave forcing.
`07_uncertainty`	Added inertia, motor mismatch, timing jitter, missed commands, watchdog, and random sensor loss.

- A cumulative comparison reveals the response of an increasingly detailed model.
  - Its last row contains every preceding effect, so the final trajectory difference is not a single-feature result.
    - An isolated comparison is the appropriate first check when asking which mechanism caused a change.

- Independent flags enable more focused studies than the preset ladder.
  - For example, retain the nonlinear M200 lookup while disabling only motor lag, or increase GNSS noise while retaining fixed latency.
    - The numerical values and command tape must remain matched to interpret the resulting difference.

4.3 Complete Boolean flag reference

The tables below specify the enabled behavior. Baseline means `00_linear`; only the three sensor-enable flags and perfect-state feedback are true in that baseline.

Hull switch	Enabled behavior
`nonlinear_kinematics`	Use the actual heading rotation instead of its local first-order approximation.
`nonlinear_coupling`	Retain velocity-product coupling instead of its tangent at the operating speed.
`nonlinear_drag`	Retain quadratic surge drag and any configured quadratic sway/yaw damping.
`added_mass`	Add separately scaled surge, sway, and yaw inertia and their consistent coupling.

Propulsion switch	Enabled behavior
`nonlinear_thrust`	Use the bounded, signed installed-scaled M200 lookup. Requires saturation.
`thrust_saturation`	Clip force to the selected forward/reverse bounds. Input percentages always stay within −100% through +100%.
`neutral_deadband`	Set requested force to zero for commands from −6.25% through +6.25%.
`reverse_asymmetry`	Use the weaker reverse branch. With a linear actuator, scale negative force by the reverse/forward limit ratio.
`motor_lag`	Apply first-order force response using the configured time constant.
`motor_rate_limit`	Limit force change per second; combine with lag when both are enabled.
`motor_mismatch`	Apply separate motor gains and retain force limits when saturation is enabled.

Timing switch	Enabled behavior
`command_sampling`	Read the input program or call a policy on periodic command ticks.
`command_delay`	Queue computation, transport, and actuation dead time before applying the command.
`timing_jitter`	Add bounded nonnegative delay. Requires command sampling and delay.
`missed_commands`	Drop individual command updates with the selected probability. Requires sampling.
`command_watchdog`	Request neutral after no command has been applied for the timeout. Requires sampling.

Environmental switch	Enabled behavior
`wind`	Apply projected-area aerodynamic force from the configured world-frame wind.
`gusts`	Add a seeded correlated wind-velocity process. Requires wind.
`apparent_wind`	Subtract boat translational velocity before evaluating aerodynamic loading. Requires wind.
`wind_yaw_moment`	Apply wind force at an offset to obtain yaw moment. Requires wind.
`wave_forcing`	Add prescribed sinusoidal planar force/moment. Independent of wind.

Sensor / observation switch	Enabled behavior
`sensor_noise`	Add separate GNSS, inertial, pixel-center, and disparity noise streams.
`sensor_bias`	Add configured GNSS position offsets and gyro bias.
`sensor_delay`	Queue measurements until their sensor availability times.
`sensor_dropout`	Invalidate randomly selected samples using separate probabilities per sensor.
`antenna_lever_arm`	Include the GNSS antenna offset in its position and velocity.
`camera_occlusion`	Invalidate target observations during the configured time interval.
`gnss_enabled`	Schedule GNSS acquisition and delivery. True in the baseline.

Sensor / observation switch	Enabled behavior
`imu_enabled`	Schedule inertial acquisition and delivery. True in the baseline.
`camera_enabled`	Schedule camera acquisition and delivery. True in the baseline.
`perfect_state_feedback`	Expose a separate exact state tuple to a policy. True in the baseline; false from the sensor preset onward.

4.4 Useful numerical controls

With the enable switches defined, numerical overrides allow sensitivity studies without editing the implementation. The values below are default test settings; they are not a component acceptance specification.

Group	Principal controls	Default
Run	Duration / integration step / log interval.	12 s / 5 ms / 20 ms.
Operating point	Forward speed / heading.	1 m/s / 0 rad.
Hull	Mass, inertia, and each drag scale.	1.0 each.
Added inertia	Surge / sway / yaw fractions.	0.10 / 0.50 / 0.20.
Geometry	Motor separation / GNSS forward offset.	0.75 m / −0.50 m.
Motor	Force time constant / force slew limit.	0.20 s / 80 N/s.
Command	Period / compute / transport / dead time.	20 / 5 / 10 / 20 ms.
Command failure	Maximum jitter / watchdog timeout.	3 ms / 100 ms.
Wind	Speed / direction / gust standard deviation.	5 m/s / 90° / 0.5 m/s.
Gust	Correlation time / process sample period.	1 s / 0.1 s.
Wave forcing	Surge / sway / yaw amplitude, period.	0 N / 1 N / 0.1 N m, 4 s.

- Hull controls include separate linear and quadratic surge scales, linear sway and yaw scales, optional quadratic sway/yaw damping, and added-inertia fractions.
  - Changing an inertia fraction does not change hull mass or nominal drag.
    - Changing mass scale also changes the geometric yaw-inertia proxy. Change inertia scale independently to study mass-distribution error.
- Actuator controls include individual motor gains, all timing intervals, and missed-command probability.
  - The uncertainty preset uses gains 0.95 and 1.05 and missed-command probability 0.02.
    - Other presets retain gains and probabilities but leave their corresponding effects disabled.
- Sensor controls include periods, latency, each noise magnitude, bias, dropout probability, geometry, and occlusion interval.
  - The uncertainty preset uses GNSS dropout probability 0.05 and camera dropout probability 0.10. Inertial dropout remains zero.
    - A camera target is a fixed world point with configurable east/north position. It is not a generated competition scene.

4.5 Reproduce a run and change one assumption

Run the commands from `EMCH792/projects/boat_control`. The first example repeats the report demonstration; the second makes a controlled sensor change.

```
uv sync
uv run python -m blueboat_sim compare \
  --set command_amplitude_percent=20 \
  --output build/configurable-simulator

uv run python -m blueboat_sim run --preset 05_sensors \
  --set sensors.gnss_position_sigma_m=0.10 \
  --set flags.sensor_dropout=true \
```

```
--set sensors.gnss_dropout_probability=0.05 \
--set seed=100 --output build/noisy-run
```

- Use `--set group.field=value` for a numerical or Boolean override.
  - Boolean values use JSON `true` or `false`; a textual string is rejected as a flag.
    - Unknown keys, invalid probabilities, nonfinite values, reversed occlusion intervals, and inconsistent flag dependencies are rejected before integration.
- Use a custom JSON file when several related settings change together.
  - `--config configs/simulator/05_sensors.json` supplies a complete template.
    - Repeated `--config` arguments compare custom templates in their supplied order. Duration, scenario, operating point, command amplitude, and initial state must match.
- Use a command tape when the input must be exactly specified.
  - `configs/simulator/commands/reversal.json` requests neutral, full forward, full reverse, and neutral at 0, 2, 4, and 6 s.
    - Commands start at zero, use finite port/starboard percentages within −100% to +100%, and have strictly increasing times within the run.

4.6 Outputs, auditing, and numerical limits

The command tape defines the requested excitation. Logs then expose the sequence between that request and the observed motion, so a run can be inspected rather than judged only by its final plot.

Saved file	Use
`resolved_config.json`	Reproduce all final settings after overrides.
`input_tape.json`	Inspect the common requested motor-percentage changes.
`truth.csv`	Plot time, six hull states, two actual motor forces, applied percentages, device pulse widths, and target force.
`metrics.json`	Compare scalar summaries under the declared experiment.
`result.json`	Audit physical parameters, source hash, trim, raw sensor records, command application, watchdog trips, and policy decision timestamps.
`comparison.json`	Store matched histories and GNSS records for downstream plots; each record marks whether it was delivered.
`summary.json`	Locate run directories and their metrics.

- Each output directory is named from the run name and seed.
  - Reusing the same directory replaces that run's files.
    - Use a new output root to retain an earlier experiment. Seeds support repeatability within the numerical environment; archive software versions for long-term reproduction.
- Hull states use fourth-order Runge-Kutta integration between exact event boundaries.
  - Motor force uses the analytic held-target lag/ramp solution at each stage.
    - Event times and integration step serve different purposes. Reduce the maximum step for numerical convergence while retaining the physical sensor and command periods.
- The validator caps a run at two million nominal integration steps and two million estimated scheduled events.
  - Nonfinite integrated states raise an error.
    - A successful run is still inspected for bounds, equilibrium behavior, and step-size convergence. Completion alone does not establish accuracy.

5 Experiments, Disturbances, and Validation

5.1 Experiment design and comparisons

The comparison runner applies a common command record to configurations that share duration, scenario, operating point, amplitude, and initial state. The saved experiment lasts 12 s, uses a 5 ms integration step, and begins at $(x, y, \psi, u, v, r) = (0, 0, 0, 1, 0, 0)$ in SI units. Each configuration receives the same four-event pulse-and-turn input tape, and seed 792 is used throughout.

- The tape starts at trim, raises both motor commands by 20 percentage points at $t = 2$ s, creates a differential command at $t = 4$ s, and returns to trim at $t = 6$ s.
  - Reusing the tape makes each run a controlled open-loop comparison under identical requested inputs.
  - Comparing internal motor force as well as position separates command interpretation and actuator response from hull and environmental effects.

- The cumulative mode enables successive groups of model features on top of each preceding stage.
  - The saved order is linear baseline, nonlinear hull, installed motor map, motor response, timing, sensors, disturbances, and parameter uncertainty.
  - A late-stage displacement therefore reflects the accumulated configuration; it cannot be attributed to that stage alone.
- Paired seeds preserve each named random stream across the compared configurations, while separate streams isolate gust, sensor, and timing draws.
  - This reduces avoidable randomness in the comparison, but a single seed does not characterize run-to-run variability.
  - Isolated mode adds each feature group to the same baseline, which is preferable when estimating that group's marginal effect.

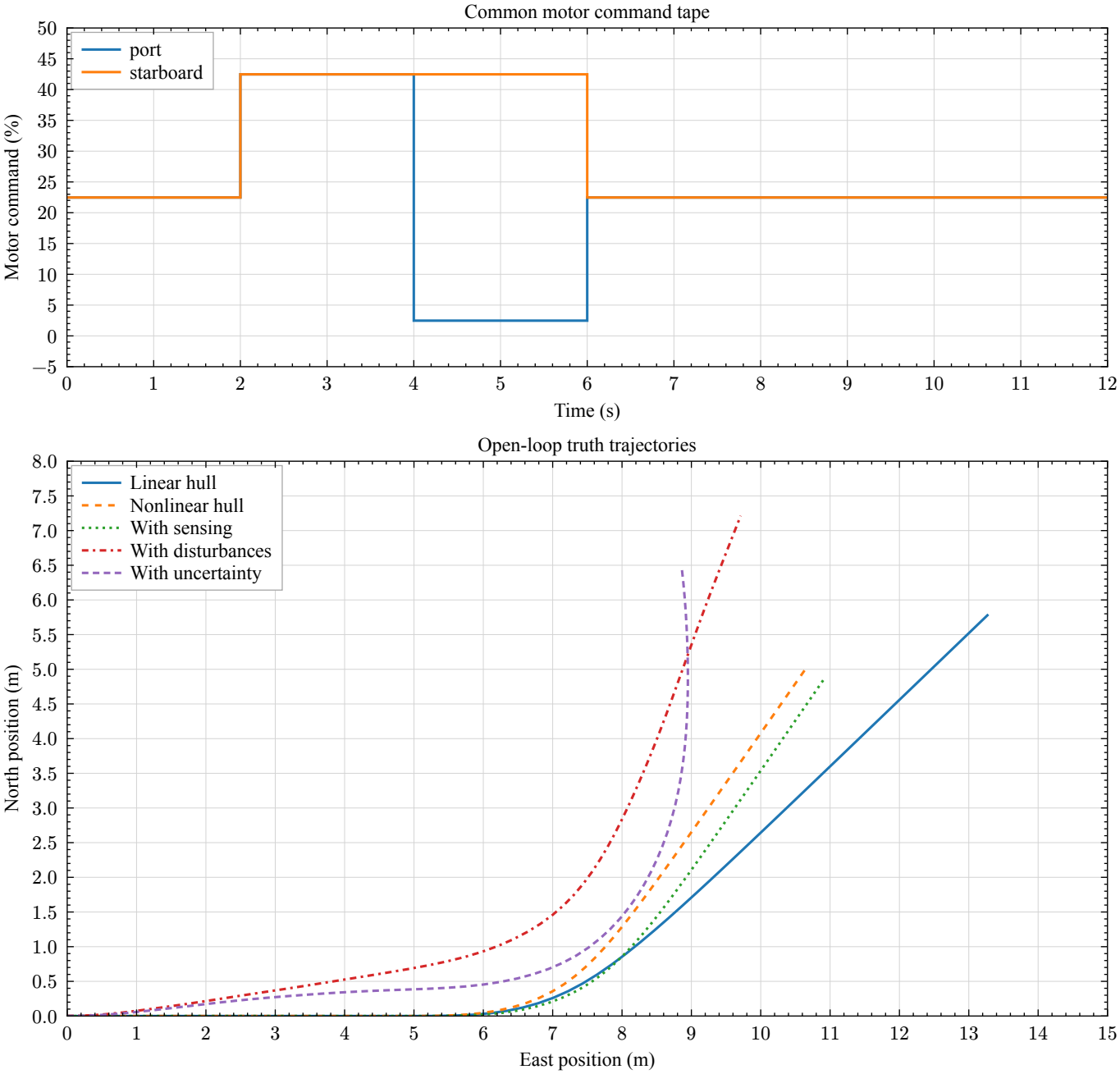


Figure 7: The common input tape and selected truth trajectories show how cumulative model stages change open-loop response.

The trajectory legend maps Linear hull to `00_linear`, Nonlinear hull to `01_hull`, With sensing to `05_sensors`, With disturbances to `06_disturbances`, and With uncertainty to `07_uncertainty`.

5.2 Wind and prescribed waves

Environmental forcing enters the planar equations as body-frame surge force X, sway force Y, and yaw moment N. The configured steady wind vector is expressed in earth axes and can include two correlated gust components. With apparent wind enabled, the vessel's earth-frame velocity is subtracted before the relative wind is rotated into body axes.

For heading ψ, earth-frame wind (w_E, w_N), and body velocities (u, v), the relative wind is $w_r^e = (w_E, w_N) - R(\psi)(u, v)$, where $R(\psi)$ rotates body components into earth axes. The model then rotates w_r^e back into body axes as (w_x, w_y). Its axis-wise projected-area drag is

$$(X_w, Y_w) = \frac{1}{2}\rho_a C_D\big(A_f w_x\ |w_x|, A_l w_y\ |w_y|\big), \tag{23}$$

where ρ_a is air density, C_D is the configured coefficient, and A_f, A_l are frontal and lateral projected areas. If the force is applied at configured body offset (x_c, y_c), its yaw moment is $N_w = x_c Y_w - y_c X_w$. The saved setting places the wind center 0.1 m forward and on the centerline, so lateral wind force produces a yaw moment while surge force does not.

- Gusts follow a first-order update for two independently driven, temporally correlated components at the configured 0.1 s interval.
  - For interval Δt and correlation time τ_g, $g_{\{k+1\}} = a g_k + \sigma_g\sqrt{1-a^2}\,\eta_k$, with $a = \exp\left(-\Delta\frac{t}{\tau_g}\right)$ and independent unit-normal two-axis draw η_k.
  - Linear interpolation between update times makes the applied wind continuous in time; it does not make the gust process a measured atmospheric record.
- The wave option adds sinusoidal components $\left[X_q, Y_q, N_q\right] = [A_X, A_Y, A_N]\sin\left(2\pi\frac{t}{T_q}\right)$ directly to the three generalized forces.
  - In the saved disturbance stage, the wave amplitudes are 0 N surge, 1 N sway, and 0.1 N m yaw with a 4 s period.
  - This prescribed planar forcing omits wave propagation, encounter frequency, heave, roll, pitch, and hydrodynamic radiation or diffraction, so it is a repeatable disturbance input rather than a sea-state simulation.

The feature set therefore supports repeatable sensitivity experiments. Its wind drag is an engineering approximation and its wave term is prescribed; neither establishes disturbance fidelity for a particular field condition.

5.3 Reading the saved run

The input tape produces clear cumulative changes in the east-north truth paths. The baseline travels 13.28 m east and 5.79 m north by 12 s. The nonlinear-hull stage ends at $(10.65, 5.01)$ m, showing that the same motor-percentage tape can produce a different path when the dynamics change. The sensor stage overlays the timing-stage truth, while prescribed disturbances move the endpoint to $(9.71, 7.21)$ m and uncertainty moves it to $(8.86, 6.46)$ m. Those endpoints describe the supplied experiment and are not evidence of improved tracking or a better controller.

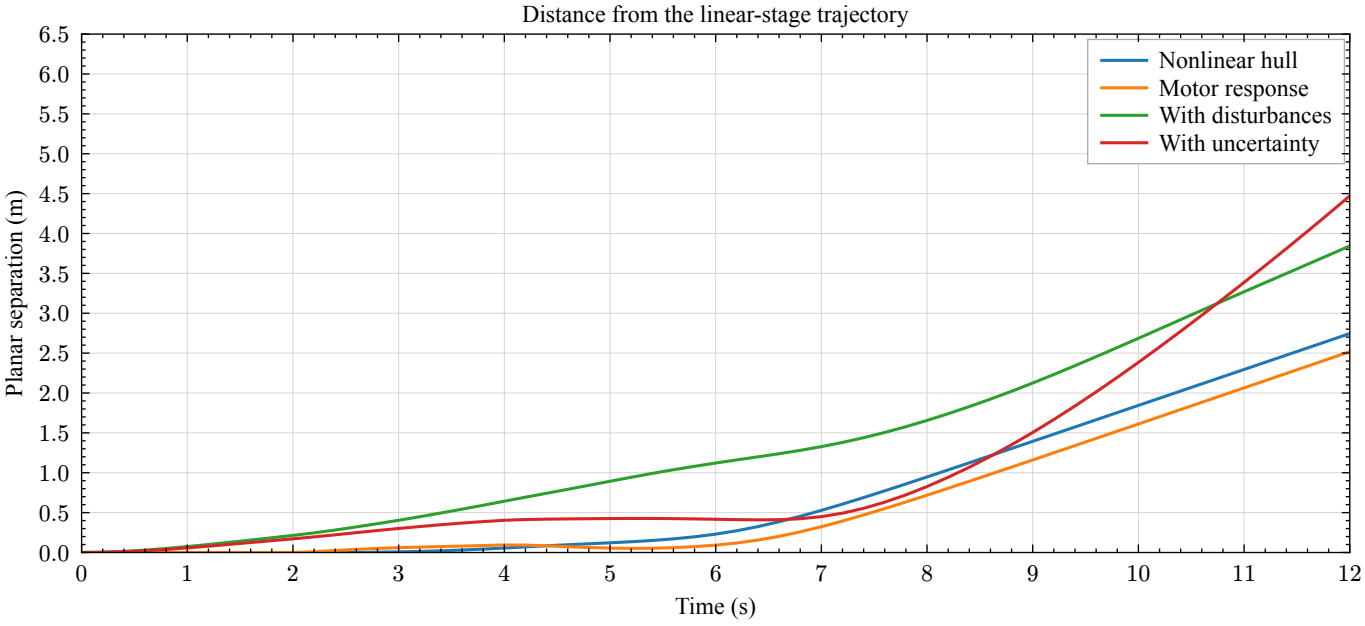


Figure 8: Selected cumulative-stage separation from the linear baseline under the same open-loop trial.

The following values are read from the saved comparison record; the force integral is the time integral of the sum of squared port and starboard thrust, not mechanical work or energy.

Stage	Final E (m)	Final N (m)	Peak (N)	∫ |T|² dt (N² s)	GNSS RMS (m)
00 linear	13.28	5.79	14.69	2080	0.000
01 hull	10.65	5.01	14.69	2080	0.000
02 motor map	10.82	5.08	15.46	2219	0.000
03 motor response	10.92	4.92	15.46	2168	0.000
04 timing	10.94	4.89	15.46	2168	0.000
05 sensors	10.94	4.89	15.46	2168	0.029
06 disturbances	9.71	7.21	15.46	2168	0.029
07 uncertainty	8.86	6.46	16.24	2224	0.029

In this table, T is the two-element vector of motor thrusts, and ∫ |T|² dt has units N² s. It is a command-effort proxy useful for comparing force histories; calling it energy would require multiplying force by velocity in a physically consistent power calculation. The sensor stage introduces a GNSS position RMS error of 0.029 m while leaving truth nearly coincident with the timing-stage path. Sensor noise, bias, delay, dropout, and acquisition latency alter delivered observations and their scoring against truth; they do not feed back into the open-loop state integration when the applied command tape is fixed.

5.4 Validation scope and next experiments

The existing tests check implementation behavior. They verify deterministic random streams, force and motor calculations, command timing, sensor causality, feature isolation, and the invariant that sensor errors do not change open-loop truth. These checks establish software properties for the implemented equations and scheduling logic. They do not measure agreement with a physical boat.

- The reported GNSS position error compares the delivered measurement's position with the truth at that measurement's acquisition time.
  - The truth point is the GNSS antenna location, including the configured lever arm, rather than the vessel reference origin.
  - Delivery latency determines when a controller could receive the record; it does not change the acquisition timestamp used for measurement error.
- A field-oriented identification study should collect synchronized PWM, GNSS, inertial, and motor or current records under documented loading and environmental conditions.
  - Estimate surge, sway, yaw, motor-map, lag, and rate parameters from selected maneuvers, then freeze those values before evaluation.
  - Hold out complete sessions, routes, and environmental conditions so adjacent samples from one maneuver do not leak across fit and test partitions.
- Future closed-loop controller comparisons should report tracking error, cross-track error, heading error, settling and overshoot, actuator saturation, force effort, and robustness across held-out seeds and conditions.
  - Score navigation against acquisition-time truth and separately report observation age and dropout so estimation error and communications delay remain distinguishable.
  - These are proposed validation measures; the saved runs do not compare controllers or demonstrate performance gains.

> The current evidence supports reproducible software tests and controlled model comparisons. Physical parameter identification and held-out boat trials are required to establish predictive accuracy.

6 Complete parameter reference

The flag catalog selects behavior, while this reference records every non-Boolean field in the baseline JSON template. Values are generated directly from that template to keep the report aligned with its runnable configuration. Field names preserve the command-line spelling. Numerical values are rounded to six decimal places here; the saved JSON retains their full precision.

- The baseline keeps most physical effects disabled while retaining their numerical settings.
  - Enabling a flag activates its associated values without replacing them.
    - The cumulative uncertainty preset changes three probabilities. Missed commands become 0.02, GNSS dropout becomes 0.05, and camera dropout becomes 0.10. It also disables perfect-state feedback as inherited from the sensor stage.
- Each group below supplies the prefix used in `--set group.field=value`.
  - Top-level experiment settings use `--set field=value`.
    - The initial-state sequence is ordered x, y, ψ, u, v, r. A custom initial velocity does not automatically change the operating-point trim used to initialize motor force.
- Camera geometry and measurement-error fields specify this simulator abstraction.
  - They do not replace camera calibration, noise characterization, or receiver status logs.
    - Delay, noise, bias, dropout, and disturbance magnitudes become physical claims only after identification under documented conditions.

6.1 Experiment settings

Field	Default value	Unit / kind
`name`	00_linear	dimensionless / identifier
`duration_s`	12	s
`integration_step_s`	0.005	s
`log_period_s`	0.02	s
`seed`	792	dimensionless / identifier
`operating_speed_m_s`	1	m/s
`operating_heading_rad`	0	rad
`initial_state`	0, 0, 0, 1, 0, 0	m, m, rad, m/s, m/s, rad/s
`scenario`	turn	dimensionless / identifier
`command_amplitude_percent`	2.5	percentage points

6.2 Hull parameters

Field	Default value	Unit / kind
`mass_scale`	1	dimensionless / identifier
`inertia_scale`	1	dimensionless / identifier
`thruster_spacing_m`	0.75	m
`surge_linear_scale`	1	dimensionless / identifier
`surge_quadratic_scale`	1	dimensionless / identifier
`sway_linear_scale`	1	dimensionless / identifier
`yaw_linear_scale`	1	dimensionless / identifier
`sway_quadratic_N_s2_m2`	0	N s²/m²
`yaw_quadratic_N_m_s2`	0	N m s²
`added_surge_fraction`	0.1	dimensionless / identifier
`added_sway_fraction`	0.5	dimensionless / identifier
`added_yaw_fraction`	0.2	dimensionless / identifier

6.3 Actuator and command parameters

Field	Default value	Unit / kind
`lag_s`	0.2	s
`rate_limit_N_s`	80	N/s

Field	Default value	Unit / kind
port_gain	0.95	dimensionless / identifier
starboard_gain	1.05	dimensionless / identifier
command_period_s	0.02	s
compute_delay_s	0.005	s
transport_delay_s	0.01	s
dead_time_s	0.02	s
jitter_max_s	0.003	s
missed_command_probability	0	dimensionless / identifier
command_timeout_s	0.1	s

6.4 Environmental parameters

Field	Default value	Unit / kind
wind_speed_m_s	5	m/s
wind_direction_deg	90	deg
wind_drag_coefficient	1	dimensionless / identifier
frontal_area_m2	0.18	m²
lateral_area_m2	0.3	m²
wind_center_x_m	0.1	m
wind_center_y_m	0	m
gust_sigma_m_s	0.5	m/s
gust_correlation_s	1	s
gust_period_s	0.1	s
wave_surge_N	0	N
wave_sway_N	1	N
wave_yaw_N_m	0.1	N m
wave_period_s	4	s

6.5 Sensor parameters

Field	Default value	Unit / kind
gnss_period_s	0.05	s
imu_period_s	0.005	s
camera_period_s	0.033333	s
gnss_delay_s	0.08	s
imu_delay_s	0.005	s
camera_delay_s	0.03	s
gnss_position_sigma_m	0.02	m
gnss_velocity_sigma_m_s	0.03	m/s
gnss_heading_sigma_rad	0.002618	rad
gyro_sigma_rad_s	0.002	rad/s
acceleration_sigma_m_s2	0.02	m/s²
gyro_bias_rad_s	0.001	rad/s
gnss_bias_e_m	0	m
gnss_bias_n_m	0	m
gnss_dropout_probability	0	dimensionless / identifier
imu_dropout_probability	0	dimensionless / identifier
camera_dropout_probability	0	dimensionless / identifier
gnss_x_m	−0.5	m

Field	Default value	Unit / kind
camera_fx_px	700	px
camera_width_px	1920	px
camera_baseline_m	0.12	m
camera_disparity_sigma_px	0.5	px
camera_center_sigma_px	2	px
camera_max_range_m	20	m
occlusion_start_s	5	s
occlusion_end_s	6	s
target_e_m	15	m
target_n_m	3	m

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